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Studierichting: Informatica
Oefeningenreeks 3G nr. 15.2

$$f(x) = 2 \sin x - \sin^2 x$$

Periode: $2\pi \Rightarrow$ onderzoek $[-\pi, \pi]$

Domein: $\mathbb{R}$

Asymptoten: geen

Nulpunten $f(x)$:

$$2 \sin x - \sin^2 x = 0$$

$$\sin x(2 - \sin x) = 0$$

$$\sin x = 0 \Rightarrow x = -\pi, 0, \pi$$

$$f'(x) = 2 \cos x - 2 \sin x \cos x$$

$$= 2 \cos x - \sin 2x$$

Nulpunten $f'(x)$:

$$2 \cos x - 2 \sin x \cos x = 0$$

$$2 \cos x(1 - \sin x) = 0$$

$$\cos x = 0 \Rightarrow x = -\frac{\pi}{2}, \frac{\pi}{2}$$

$$\sin x = 1 \Rightarrow x = \frac{\pi}{2}$$

$$\Rightarrow x = -\frac{\pi}{2}, \frac{\pi}{2}$$

$$f''(x) = -2 \sin x - 2 \cos 2x$$

Nulpunten $f''(x)$:

$$-2 \sin x - 2 \cos 2x = 0$$

$$\sin x + \cos 2x = 0$$

$$\sin x + 1 - 2 \sin^2 x = 0$$

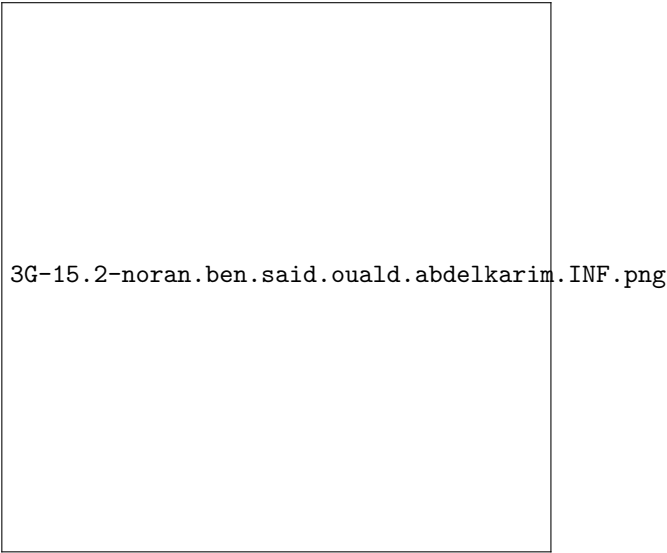
$$2 \sin^2 x - \sin x - 1 = 0$$

$$(2 \sin x + 1)(\sin x - 1) = 0$$

$$\sin x = -\frac{1}{2} \Rightarrow x = -\frac{5\pi}{6}, -\frac{\pi}{6}$$

$$\sin x = 1 \Rightarrow x = \frac{\pi}{2}$$

	$-\pi$	$-\frac{5\pi}{6}$	$-\frac{\pi}{2}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{2}$	π			
$f(x)$	0	−	−	−	−	0	+	+	0	
$f'(x)$		−	−	0	+	+	+	0	−	
$f''(x)$		−	0	+	+	0	−	−	0	
$f(x)$		$\searrow$	$\searrow$	$m(-3)$	$\nearrow$	$\nearrow$	$\nearrow$	$M(1)$	$\searrow$	
$f(x)$		$\frown$	$B\left(-\frac{5}{4}\right)$	$\smile$	$\smile$	$B\left(-\frac{5}{4}\right)$	$\frown$	$\frown$	$B(1)$	$\frown$



References

[1] Grafiek door desmos